The force experienced by north pole,

$$\vec{F}_N = q_m \vec{B} \qquad (3.23)$$

The force experienced by south pole,

$$\vec{F}_S = -q_m \vec{B} \qquad (3.24)$$

Adding equations (3.23) and (3.24), we get the net force acting on the dipole as

$$\vec{F} = \vec{F}_N + \vec{F}_S = \vec{0}$$

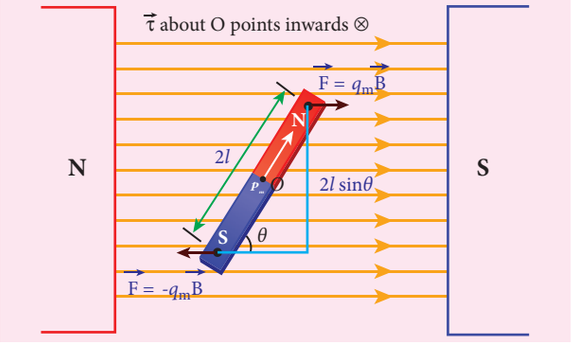


Figure 3.16 Magnetic dipole kept in a uniform magnetic field

The moment of force or torque experienced by north and south pole about point O is

$$\vec{\tau} = \overrightarrow{ON} \times \vec{F}_N + \overrightarrow{OS} \times \vec{F}_S$$
$$\vec{\tau} = \overrightarrow{ON} \times q_m \vec{B} + \overrightarrow{OS} \times \left(-q_m \vec{B}\right)$$

By using right hand cork screw rule, we conclude that the total torque is pointing into the paper. Since the magnitudes $\left|\overrightarrow{ON}\right| = \left|\overrightarrow{OS}\right| = l$ and $\left|q_m \vec{B}\right| = \left|-q_m \vec{B}\right|$, the magnitude of total torque about point O

$$\tau = l \times q_m B \sin\theta + l \times q_m B \sin\theta$$
$$= 2l \times q_m B \sin\theta$$
$$\tau = p_m B \sin\theta \qquad \left(\because q_m \times 2l = p_m\right)$$

In vector notation, $\vec{\tau} = \vec{p}_m \times \vec{B}$ (3.25)

DO YOU KNOW?

(a) Why a freely suspended bar magnet in your laboratory experiences only torque (rotational motion) but not any translatory motion even though Earth has non-uniform magnetic field?

It is because Earth's magnetic field is locally(physics laboratory)uniform.

(b) Suppose we keep a freely suspended bar magnet in a non-uniform magnetic field. What will happen?

It will undergo translatory motion (net force) and rotational motion (torque).

3.3.1. Potential energy of a bar magnet in a uniform magnetic field

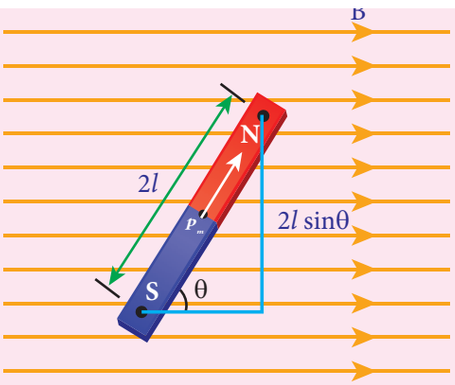


Figure 3.17: A bar magnet (magnetic dipole) in a uniform magnetic field

When a bar magnet (magnetic dipole) of dipole moment $\vec{p}_m$ is held at an angle θ with the direction of a uniform magnetic field $\vec{B}$, as shown in Figure 3.17 the magnitude of the torque acting on the dipole is

$$\left|\vec{\tau}_B\right| = \left|\vec{p}_m\right|\left|\vec{B}\right|\sin\theta$$

If the dipole is rotated through a very small angular displacement $d\theta$ against the

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